

Learning the Readout Form: Objective-Level Calibration, Sparse Selectivity, and Expert Routing on a Diagonal State-Space Host

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Abstract

Online adaptation on a state-space host is often judged with a human-specified output form. We show that this choice is load-bearing. On the same diagonal SSM host and paired stream protocol as Paper D, a squared-loss RLS readout evaluated with a clip floor parks 1.6–10.2% of tokens on a fixed floor; several stream comparisons become floor-sensitive. Replacing the evaluation—not the numeric range—with a trained softmax readout removes the floor and lowers stream cross-entropy by 0.307 nats/token (perplexity $11.75 \rightarrow 8.65$, 10/10 seeds), at a retention cost of +0.32 same-token-set ppl (1/10). A post-hoc simplex projection that removes all negatives does *not* repair the metric (stream Δ ppl +0.03, 0/10): the artifact lives in the objective, not the range. Under the clean metric the ungated full state remains stream-neutral and retention-harmful; a 128-dimensional noise control rules out dimension/optimizer imbalance, while a 32-dimensional state slice is stream-positive. Unconstrained cross-entropy training of a feature-level gate improves stream form but does not sparsify and destroys retention. Hard top- k sparsity ($k = 32$) is the missing pressure: it is the best stream arm that keeps retention near baseline (7.03; a weakly L_1 -regularised gate edges it on stream, 6.99, at 11.57 retention). Soft routing over two online softmax experts reduces forgetting further ($9.20 \rightarrow 7.79$, 10/10) at stream parity; a hard-routing ablation is slightly better on retention (6.94) at a small stream tax. The full stack beats the calibrated input-path baseline on both axes. We state a design rule—the evaluation instrument must be part of the learned system—and the

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boundary where it stops paying: candidate-family breadth remains human-supplied. All results are CPU-scale; we make no scaling or state-of-the-art claims.

Keywords: online learning, state-space models, readout calibration, selective state space, continual learning, falsification

1. Introduction

Self-correcting learning systems are usually evaluated with instruments the designer fixed in advance: a loss, a readout parameterization, a scoring rule. When those instruments are part of what training changes, evaluation and improvement can share a gradient. When they are frozen and human-prescribed, the system optimizes a different quantity than the one used to judge it, and artifacts of that mismatch can survive any amount of downstream “self-correction.”

Paper D (companion; [1]) diagnosed a diagonal state-space host under such a prescribed instrument: per-token recursive least squares (RLS) on a linear readout, scored as $CE = -\ln \text{clip}(\hat{y}_t[e_t], 10^{-12}, 1)$. That work established a deconvolution impossibility for recovering the current token from a linearly decayed state mixture, identified the additive input path as required for calibration, and showed that several stream rankings reverse once clipped tokens are excluded. Paper E [2], on a different substrate, measured the complementary ceiling at the decision layer: under a ± 1 reward channel, autonomy reduces to selection among a human-supplied pair of correction laws.

This paper is written *as the rule* extracted from those traps. We keep Paper D’s host, task, and seed rules so that every row pairs bitwise with the diagnostics, and we change what is allowed to be learned:

1. **C1 (objective, not range).** A trained softmax readout removes the clip floor and lowers stream CE by 0.307 nats/token (ppl -26%), with an honest retention cost in the same sentence.
2. **C2 (post-hoc control).** Projecting the same RLS solution onto the probability simplex after the fact eliminates all negatives and does *not* improve stream ppl (0/10). Range fixes are decoration.
3. **C3 (boundary under a clean metric).** The ungated full state is stream-neutral and retention-harmful; dimension and optimizer imbalance are excluded by noise and width-matched controls.

4. **C4/C4b (learned sparse selectivity)**. Unconstrained cross-entropy training of a feature-level gate improves stream form but does not sparsify and collapses retention. Hard top- k sparsity is the missing pressure: best stream among the retention-safe gate variants.
5. **C5 (expert routing)**. Soft routing over two online softmax experts on the sparse-gated base reduces forgetting by 1.41 ppl (10/10) at stream parity.

Feedback budget. Following the reporting discipline of Papers D and E, every autonomy or form-discovery claim is stated together with the information the system was allowed to see. Here the budget is *per-token labels* via cross-entropy—strictly richer than Paper E’s ± 1 channel. We do not port Paper E’s decision-layer mechanism stack onto this host: the reason those mechanisms exist (no gradient to a parameter) is absent once a calibrated CE objective is available.

Claims we do not make. Proposition 1 of Paper D is not “beaten”; our gains come from recalibrating the input path and from sparse gating, not from inverting a linear deconvolution. We do not claim SOTA against Mamba/TTT/DeltaNet, scaling, or multi-step RL. All results are CPU-scale ($N = 128$) on a two-domain biased-bigram stream.

2. Related work

Online learning inside sequence models. Test-time training and related ideas update hidden representations or readouts at inference [3]. Linear-recurrence and selective SSM architectures provide a host whose state can be updated online [3, 4, 5]; fixed random reservoirs with trained readouts are the classical online-readout substrate [6, 7]. Paper D positioned an explicit per-token RLS readout and REDEM mechanism division of labor against amortized online learners and surprise-based memories; parameter-efficient adapters such as LoRA [8] are the usual amortized reference. We inherit the host and change only the readout form and selectivity pressure.

Metric artifacts in online CE. Clipped linear scores on squared-loss readouts create a floor mass that can drive stream rankings. Paper D reported clip fractions and unclipped sensitivity analyses. This paper adds the missing control: a post-hoc range fix (simplex) versus an objective fix (trained softmax).

Sparse and gated features.. Hard cardinality constraints (top- k) and ℓ_1 pressures are standard tools for feature selection. Our use is online and coupled to a recurrent host: the gate multiplies whitened state channels while the current-token input path stays clean—the additive path that Paper D showed is required.

Mixture of experts online.. Soft mixtures of specialists with responsibility-weighted updates are classical. Here experts are *online softmax readouts* selected by a fast-channel EMA domain statistic (Paper D’s M3 metadata), under a clean metric, on a sparse-gated state slice. Catastrophic-forgetting baselines such as EWC [9] and consolidation accounts of multi-timescale memory [10] are not the focus here (Paper E [2] reports a matched-optimizer negative); we stay inside the readout-form question.

3. Preliminaries: host, metric, and Paper D’s boundary

3.1. Diagonal SSM host

Following Paper D,

$$h_t = A \odot h_{t-1} + B e_{t-1}, \quad A = \text{diag}\left(\exp(-1/\tau_i)\right), \quad (1)$$

with log-uniform $\tau_i \in [1, 3000]$ and spectrum whitening $h_t^w = h_t \odot \sqrt{N(1 - A_i^2)}$, $N = 128$. The additive input path $x_t = B e_{t-1}$ is retained as a clean feature (D: the current token cannot be recovered from the linearly mixed state by a linear readout).

3.2. Clipped-linear metric (Paper D)

Squared-loss RLS on ϕ produced $\hat{y} = W\phi$, scored as

$$\text{CE} = -\ln \text{clip}(\hat{y}[e_t], 10^{-12}, 1), \quad (2)$$

The input-path arm parks $\sim 1.6\%$ of stream tokens on the floor; state-feature arms park $\sim 10\%$. Paper D showed rankings that reverse under unclipped sensitivity metrics and scoped several stream claims as floor-sensitive.

3.3. What this paper changes

We treat the readout *map* and the *gate form* as trainable objects under a proper probabilistic objective, and we ship the post-hoc simplex arm next to every headline (Section 5).

4. Method

4.1. C1: trained probabilistic readout

Let ϕ_t be any feature concatenation ending with a bias. We predict

$$p_t = \text{softmax}(W\phi_t), \quad \mathcal{L}_t = -\ln p_t[e_t], \quad (3)$$

and update W with AdaGrad (lr 0.05, gradient clip 5.0; plain SGD exploded on high-dimensional skip features). Softmax outputs are strictly positive; `floor_frac = 0` is true by construction and is never quoted as an empirical finding.

4.2. C2: post-hoc simplex control

As a control, we keep RLS training on squared error and only change the evaluation map to $\Pi_\Delta(\hat{y}) = \max(\hat{y}, 0) / \sum_j \max(\hat{y}_j, 0)$ (uniform fallback if the mass vanishes). This is clip-negatives plus renormalize — not a Euclidean projection onto the probability simplex — and it removes every negative score without changing the learned W .

4.3. C4b: sparse learned gate

$$g_t = \text{TopK}_k(\sigma(Cx_t + b)), \quad \phi_t = [h_t^w \odot g_t; x_t; 1], \quad (4)$$

with $k = 32$ matched to the width of the best fixed state slice. C, b are trained jointly with the readout under the same CE objective; non-selected channels receive zero gradient through the gate (straight through the hard mask). An ℓ_1 variant is reported as an ablation.

4.4. C5: soft expert routing

Fast-channel EMA metadata (Paper D M3; $\tau_i \leq 8$, $\tau_m = 500$) yields m_t and fixed per-domain references. Soft routing uses

$$\pi_{t,k} = \text{softmax}_k(-\|m_t - \text{ref}_k\|/T_r), \quad p_t = \sum_k \pi_{t,k} \text{softmax}(W_k\phi_t), \quad (5)$$

with exact mixture responsibility $\pi_{t,k} p_{t,k}[e_t]/p_t[e_t]$ scaling each expert’s CE gradient. Hard nearest-reference routing (Paper D A3 semantics) is an ablation. Holdout evaluation of each expert uses a single-step EMA of the matching domain reference (same metadata path as stream); this is disclosed here because it injects domain identity at evaluation time and is not a fully blind holdout.

5. Experiments

5.1. Protocol

Task and seeds are verbatim from Paper D / s18: two biased-bigram Markov domains alternating every 3000 tokens, 6 segments, 10 seeds, seed rule $\text{seed} \times 101 + 17$. The control arm **B-lin-clip** reproduces Paper D’s **B-proj** stream, forgetting, and clip fraction exactly ($\max |\Delta| = 0$ on all three metrics, 10/10). Forgetting is the held-out previous-domain perplexity of the final readout (domain-matched specialist when routing). Head-to-head statements use same-token-set CE where relevant. We report means over seeds and sign counts; we do not merge `clip_frac` with `floor_frac`.

5.2. C1–C2: readout form

Table 1: Readout form (10-seed means). Post-hoc simplex removes negatives (`neg_frac` $\rightarrow$ 0) but does not repair stream ppl (C2). Trained softmax does (C1), with a small retention cost.

Arm	stream	forget	clip%	floor%
B-lin-clip (D B-proj)	11.754	8.387	1.58	–
B-simplex (post-hoc)	11.783	8.387	0	1.58
B-softmax-sgd	8.648	8.890	0	0 [†]
Skip-lin-clip (D CV-skip)	58.465	94.057	10.23	–
Skip-simplex	75.736	140.234	0	10.23
Skip-softmax-sgd	8.629	57.711	0	0 [†]

Same-token-set stream: B-softmax vs B-lin-clip $\Delta\text{CE} = -0.307$ (10/10; paired $t = -92.3$ from an offline seed-level reduction of the committed per-token CE arrays, not a separate script); B-simplex vs B-lin-clip $\Delta\text{ppl} = +0.03$ (0/10). Retention price of C1: same-set $\Delta\text{ppl} +0.32$ (1/10). The range-fix story is visualized in Figure 1: the post-hoc simplex relocates the clipped negative mass to the floor, while the trained softmax removes it by construction.

5.3. C3: state under a clean metric

B-softmax vs Skip-softmax: stream $\Delta\text{CE} +0.003$ ($t = 0.22$); forget $\Delta\text{CE} -1.348$ (10/10). Noise does not collapse retention (dimension is not the cause). A 32-dimensional slice is stream-positive vs B-softmax ($\Delta\text{CE} +0.061$, 9/10) with mild retention cost.

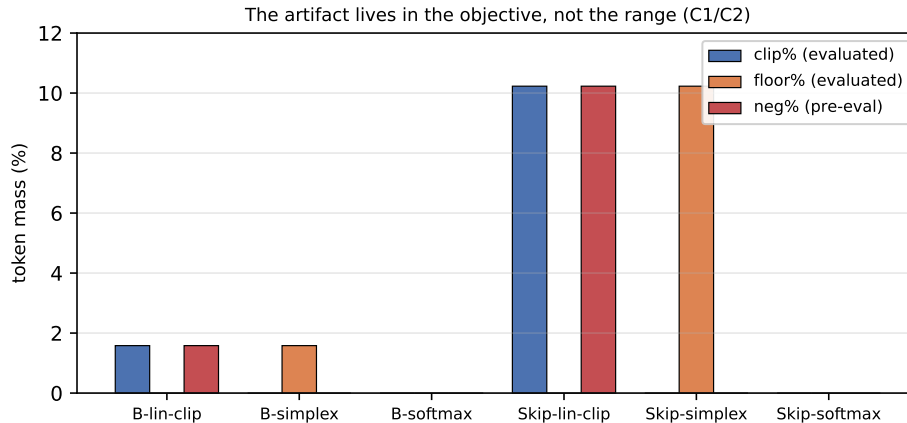


Figure 1: Clip-floor artifact across readout forms (10-seed means). Negatives under the clipped-linear metric (**neg_frac**) are parked on the floor by clipping (**clip_frac**); the post-hoc simplex arm relocates that mass to **floor_frac** without improving stream ppl (C2). The trained softmax readout has none of either by construction (C1).

Table 2: Dimension/optimizer controls under softmax (10-seed means).

Arm	stream	forget
B-softmax-sgd	8.648	8.890
B-noise-softmax (ε^{128})	10.271	9.554
Skip-sub-softmax ($h_{[:32]}^w$)	8.142	11.231
Skip-softmax-sgd (full h^w)	8.629	57.711

5.4. C4/C4b: learned selectivity and sparsity

Gate-C-topk vs B-softmax: stream better 10/10 (-1.62); forget mean $+0.31$ (3/10 better) — near parity. Gate-C-topk vs Gate-C: forget better 10/10 (-21.9). Cross-entropy alone does not sparsify; hard cardinality is the pressure that turns selectivity into a retention-safe mechanism. The weakly L_1 -regularised gate (Gate-C-L1) does sparsify — it opens only 6% of the gates above 0.5, against 25% ($= k/N$) for top- k — and edges top- k on stream by 0.04 ppl, but it pays 2.37 ppl more on retention (11.574 vs. 9.201) and 1.81 ppl more on the across-seed median (11.136 vs. 9.330). It is therefore reported for completeness and not selected, and every “best stream” statement in this paper means best among the variants that keep retention near baseline.

Table 3: Gates under CE (10-seed means). Unconstrained Gate-C is the best non-sparse stream arm but fails retention; top- k fixes both.

Arm	stream	forget mean/med	notes
Skip-sub	8.142	11.231 / 9.908	fixed slice
Gate-C	7.741	31.075 / 26.155	gate ≈ 0.72 open
Gate-C-topk	7.033	9.201 / 9.330	$k = 32$
Gate-C-L1	6.994	11.574 / 11.136	$\lambda=0.01$, 6% open

5.5. C5: expert routing

Table 4: Routing on the Gate-C-topk base, $\tau_m = 500$ (10-seed means).

Arm	stream	forget mean/med	Δ forget vs Single
Single-topk	7.033	9.201 / 9.330	—
Soft-topk	7.065	7.789 / 7.717	−1.41 (10/10)
Hard-topk	7.111	6.939 / 6.913	−2.26 (10/10)

Soft vs Single stream Δ CE mean +0.03 ($t = 0.68$) — parity. Hard pays a small stream tax for stronger retention (Paper D’s specialization–lag tradeoff survives under a calibrated readout). **Full stack vs B-softmax**: Soft-topk stream 7.065 vs 8.648; forget 7.789 vs 8.890 — both axes better. Figure 2 summarizes the headline arms across C1–C5 on both axes (10-seed means $\pm$ std).

6. Analysis

Why simplex fails and softmax works.. RLS on squared error is free to drive the target coordinate non-positive; clipping or projecting after the fact cannot restore mass that the objective never placed on the simplex. Softmax CE constrains the map during learning. C2 is therefore not a minor ablation: it is what makes C1 a statement about objectives rather than about numeric hygiene.

Why CE does not produce a sparse gate.. Stream CE is largely satisfied by a moderately open gate that keeps domain-level and mixture information; that same open gate is what poisons held-out retention (C3). Sparsity must be imposed as a constraint (top- k), not hoped for from the task loss.

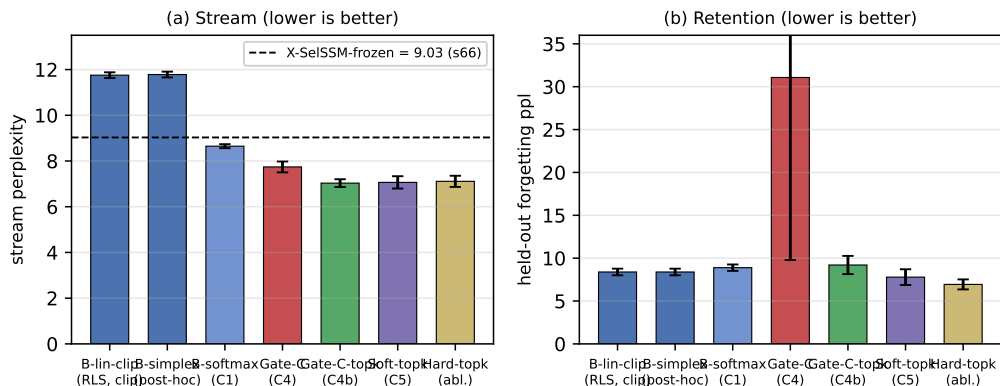


Figure 2: Headline arms across C1–C5 (10-seed means $\pm$ std). (a) Stream perplexity: the trained softmax readout (C1) removes the clip floor and improves stream; hard top- k gating (C4b) is the best stream arm and sits below the frozen selective-SSM reference (dashed line, s66). (b) Held-out forgetting: the unconstrained gate (C4) collapses retention, top- k restores parity, and soft expert routing (C5) converts the remaining stream–retention tension into a further retention gain.

Hard vs soft routing.. Hard routing purer specialists (lower forget) at a lag cost on stream; soft routing mixes and keeps stream at parity while still buying 1.4 forget. This mirrors Paper D’s soft-vs-abrupt finding under a different readout family.

Design rule.. Self-improvement requires a trustworthy self-evaluation, and the instrument must be learnable rather than prescribed. D prescribed the readout form; E prescribed correction laws; F trains the form and ships the post-hoc control that would have hidden the difference.

7. Discussion

Feedback budget.. All form-discovery results use per-token labels. They are not comparable to ± 1 decision-layer autonomy without stating that budget. We deliberately do not port Paper E’s mechanism stack: once a gradient exists, the reason for a five-mechanism rule-selection tower largely disappears, while retention must be re-measured under CE.

Boundary.. Rules that make evaluation trustworthy do not generate new candidate families. F learns form *inside* supplied families (softmax maps,

gate parameterizations, expert mixtures). The next ceiling is candidate-space *breadth*: a trainable generator of hypotheses, not only a scorer of human-supplied ones.

External baseline (s66) and the frozen-host surprise.. All s66 numbers cited below are 10-seed means over the committed rows of `data/s66_external_ssm_baseline_v1.json` (single configuration: $N=128$ diagonal selective-SSM host on the char-level biased-bigram stream; `stream_ppl` is the stream perplexity and `forgetting_ppl` the held-out previous-domain perplexity; parameter counts are the `n_params_readout/n_params_gate/ n_params_host` fields of the same file). A Mamba-style selective-SSM arm on the identical host, stream, seeds and metric does not close the gap to F’s learned top- k gate (stream 9.03 vs. 7.03, 0/10 external better), and joint online training of that external host is destructive. The more informative observation is internal to the external pair: the *frozen* selective-SSM arm is the stronger external competitor (stream 9.03 vs. 10.19; forget 10.38 vs. 115.24), yet it still loses to F’s calibrated softmax-sgd anchor on stream (9.03 vs. 8.65) and on retention (10.38 vs. 8.89). That ordering — random frozen selectivity $>$ jointly trained host — points to a pretrained-versus-random-host comparison, not to a claim that selective SSMs are inferior in general. We therefore treat s66 as a scoped external check that licenses only the selective-SSM parameterisation at this protocol scale.

Limitations.. Two-domain biased-bigram stream; primary tables at $N = 128$; one host configuration. A Mamba-style selective-SSM baseline (s66) has now been run on the same host, stream, seeds and metric: the frozen-host external arm reaches stream ppl (`stream_ppl` mean) 9.03 and the jointly trained host arm is worse on both axes (10.19 stream at the matched learning rate; retention collapses; a $4\times$ larger host learning rate makes joint training diverge, so the committed s66 rows pool both rates under one arm name), while F’s learned top- k gate remains best (7.03, 10/10 paired seeds vs. the frozen external arm). The two arms are *not* capacity-matched, and the mismatch runs in F’s favour on the axis that matters for training: the frozen external arm’s 24,864 host parameters are held at their per-seed *random* initialisation, so its trained budget is the shared readout alone (8,224), while F’s *Gate-C-topk* arm trains 24,736 (8,224 readout + 16,512 gate) — $3.0\times$ more (fields `n_params_readout/n_params_gate/ n_params_host` of the s66 rows). In total parameter count the external arm is the larger of the two

(33,088 vs. 24,736), so neither budget comparison is like-for-like and we do not claim one: the margin over the frozen arm is a positive result against a random selective-SSM feature map at this scale, not against a trained modern SSM. Against F’s ungated **B-softmax-sgd** anchor (4,128 trained) the external arm carries $2.0\times$ the trained parameters and still loses on both axes. This closes the earlier “no external modern-SSM baseline” gap for the *selective-SSM* family only — one host configuration, one protocol family, $N = 128$, 10 seeds, CPU scale; it is not a state-of-the-art comparison and does not cover DeltaNet, gated linear attention, or TTT (the TTT fast-weight arm scored worse than the uniform predictor at this protocol scale and was removed as non-viable). A width sweep ($N \in \{128, 256, 512\}$, 5 seeds) with $k = \max(8, N/4)$ showed the calibrated input-path baseline improving with N (stream $8.63 \rightarrow 7.25$) while sparse-gate and soft-routing arms failed at $N = 512$ (Gate-C-topk stream 9.16, forget 27.3). A follow-up k -sweep at $N = 512$ (3 seeds) shows this is a *sparsity* failure, not a width impossibility: $k \in \{16, 32\}$ restores the ordering (Gate-C-topk $k = 32$: stream 6.79, forget 9.89; Soft-topk $k = 16$: stream 6.59, forget 6.27) and beats the $N = 512$ input-path baseline (7.25/11.09), whereas $k = 128$ reproduces the collapse. Claims should therefore be read as requiring a *fixed* modest k (or a tuned $k(N)$), not $k \propto N$. A Mackey-Glass bin-prediction task (5 seeds) supports transfer of the sparse-gate advantage off the bigram stream (train/holdout ppl $\approx 20/16$ vs $30/29$ for the input-path baseline). Softmax `floor_frac` = 0 is structural. These remain proof-of-concept bounds.

8. Conclusion

Paper D left a prescription: a calibrated output that removes the clip floor, and learned selectivity as an open boundary. This paper ran the prescription, falsified the post-hoc range fix, and showed that the boundary only opens when selectivity is both *learned* and *sparsity-pressured*, after which expert routing converts the remaining stream-retention tension into a further retention gain. The general lesson is narrower than “use softmax” and stronger than a single mechanism: **put the evaluation instrument inside the learned system**, and report the feedback budget next to every autonomy claim.

Code and data availability

All simulation scripts and generated data reported here are openly available at <https://github.com/huyamingc/REDEM>.

The experiment scripts cited in this paper are CSVs and scripts under `data/` and `scripts/` (paired to Paper D s19/s20):

- `s50_paper_f_pilot_nl_readout` (C1–C3);
- `s51_paper_f_learned_gate` (C4/C4b);
- `s52_paper_f_soft_route_experts` (C5);
- `s53_paper_f_scaling` (width sweep);
- `s53b_paper_f_ksweep` (k -sweep at $N = 512$);
- `s54_paper_f_mackey_glass` (Mackey–Glass transfer);
- `s66_external_ssm_baseline` (selective-SSM baseline).

CRedit authorship contribution statement

Yaming Hu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing.

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During the preparation of this work the author used an AI-assisted writing and review tool in order to improve readability, check consistency and draft editorial text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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